

Graph: ggtree and its main functions

Introduction

The **ggtree** package in R is a powerful tool for visualizing and annotating phylogenetic trees. It is built on top of the **ggplot2** package and allows for highly customizable and publication-ready tree visualizations.

Prerequisites

To follow this tutorial, ensure you have the following R packages installed:

```
install.packages("tidyverse")
if (!requireNamespace("BiocManager", quietly = TRUE)) {
  install.packages("BiocManager")
}
BiocManager::install("ggtree")
BiocManager::install("treeio")
```

We will use a **.nwk** dataset for demonstration.

```
library(ggtree)
library(treeio)
library(ggplot2)
library(dplyr)
```

Loading a Phylogenetic Tree

ggtree supports various formats of phylogenetic trees. The most common is the Newick format.

```
# Load the tree from a Newick file
tree <- read.newick("example_2_phylo.nwk")

# Print tree structure
print(tree)
```

Phylogenetic tree with 110 tips and 108 internal nodes.

Tip labels:

GCF_029224125.1, GCF_024584905.1, GCF_003176855.1, GCF_009867035.2, GCF_030285565.1, GCF_0

Node labels:

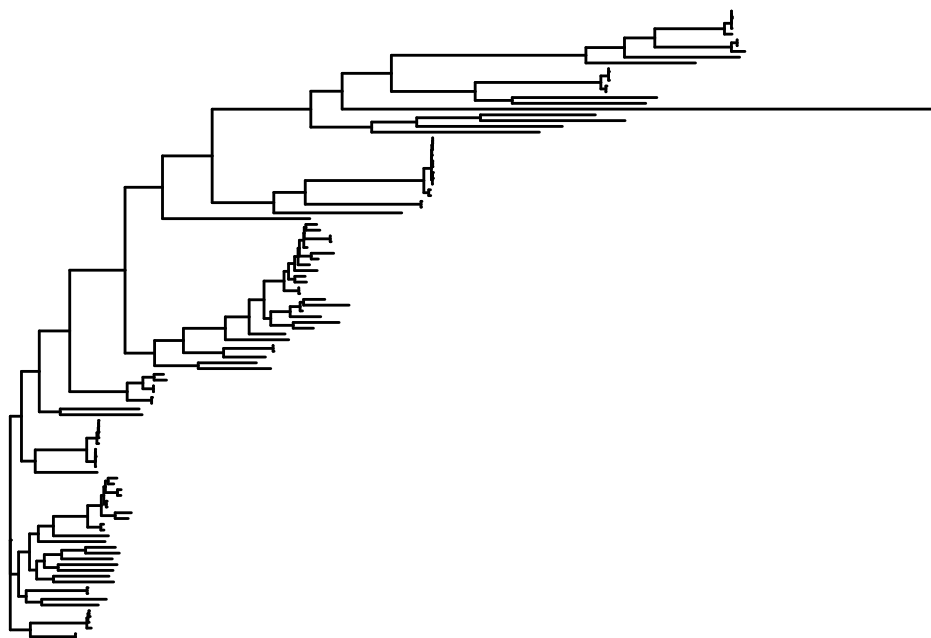
, 0.981998, 0.987349, 0.946895, 0.994059, 1.000000, ...

Unrooted; includes branch lengths.

Basic Tree Visualization

Visualizing the tree is straightforward:

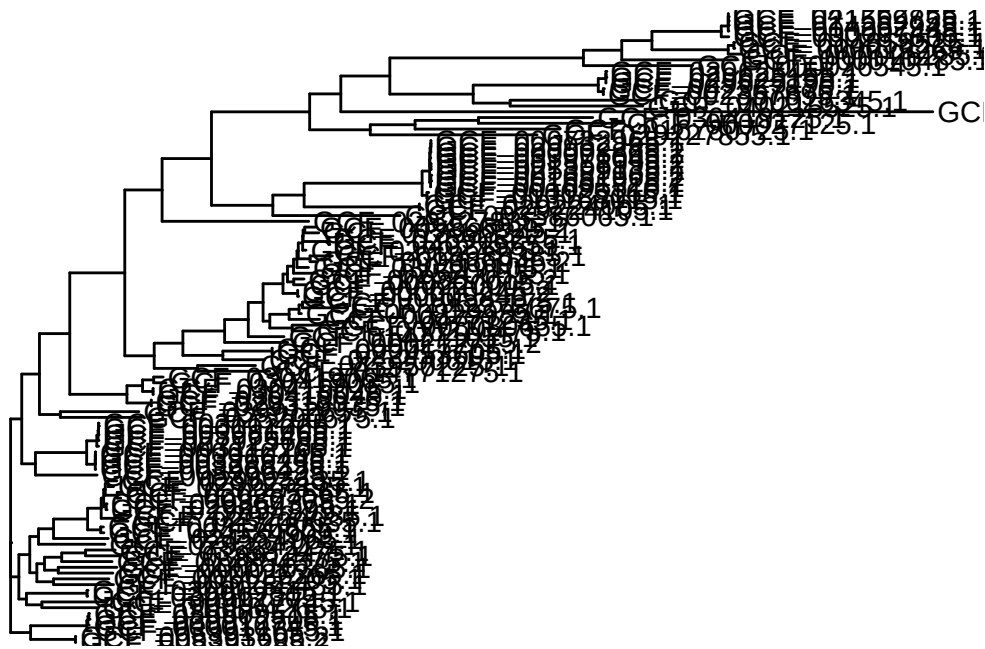
```
# Basic tree plot
p <- ggtree(tree)
print(p)
```



Adding Tip Labels

You can add labels to the tips of the tree using `geom_tiplab`:

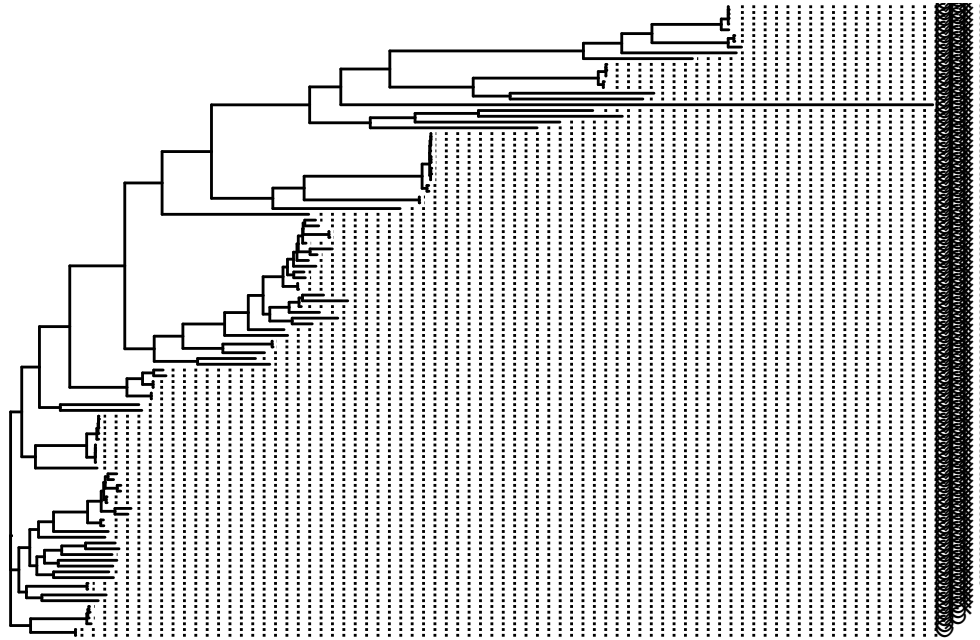
```
p <- ggtree(tree) +  
  geom_tiplab()  
print(p)
```



Parameters for `geom_tiplab`

- `size`: Adjusts the size of the labels.
- `align`: Aligns labels horizontally.
- `angle`: Rotates labels (useful for circular trees).

```
p <- ggtree(tree) +
  geom_tiplab(size = 3, align = TRUE, angle = 45)
print(p)
```

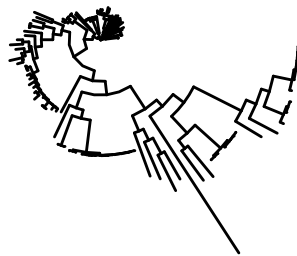


Tree Layouts

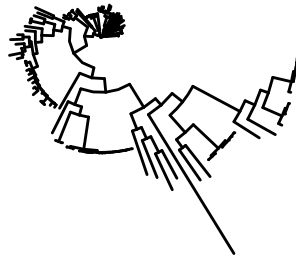
ggtree supports various layouts for trees, such as:

- rectangular (default)
- circular
- fan
- slanted
- radial

```
# Circular layout  
p_circular <- ggtree(tree, layout = "circular")  
print(p_circular)
```



```
# Fan layout
p_fan <- ggtree(tree, layout = "fan")
print(p_fan)
```



Annotating Trees

Adding Node Labels

To annotate nodes, use `geom_nodelab`:

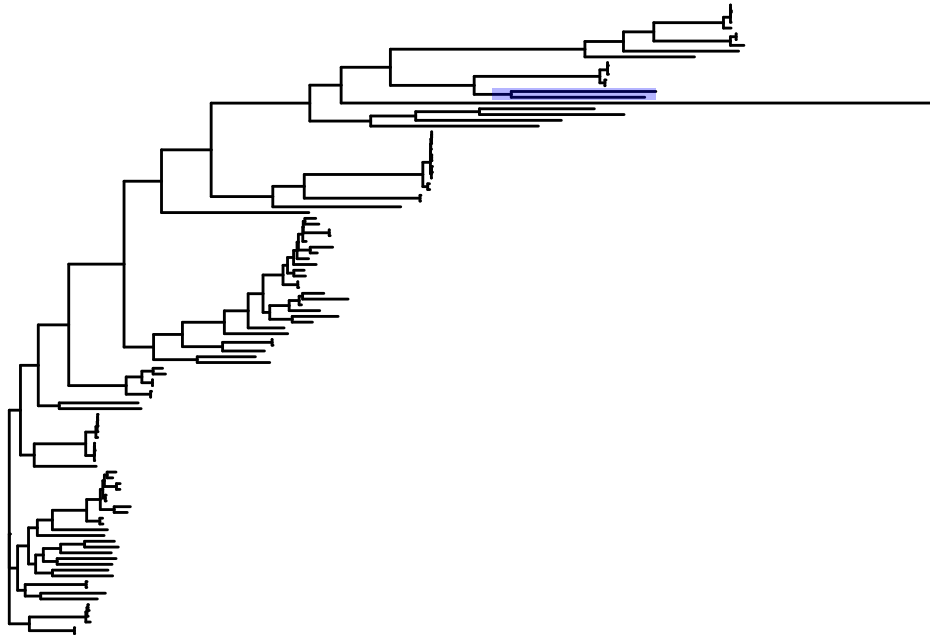
```
p <- ggtree(tree) +
  geom_nodelab(aes(label = node), size = 3)
print(p)
```



Highlighting Clades

Highlight specific clades with `geom_highlight`:

```
p <- ggtree(tree) +  
  geom_highlight(node = 199, fill = "blue", alpha = 0.3)  
print(p)
```



Parameters for `geom_highlight`

- `node`: Specify the node ID to highlight.
- `fill`: The fill color for the highlight.
- `alpha`: Transparency of the highlight.

Annotating Specific Tips

```
p <- ggtree(tree) +  
  geom_tippoint(aes(color = label), size = 3)  
print(p)
```

Adding Metadata

Metadata can be merged into the tree object and displayed, metadata is a `data.frame`:

```
# Cargar metadata
metadata <- readr::read_tsv("Experiment_Design_example_2.tsv")
metadata <- metadata %>%
  rename(label = Assembly.Accession)

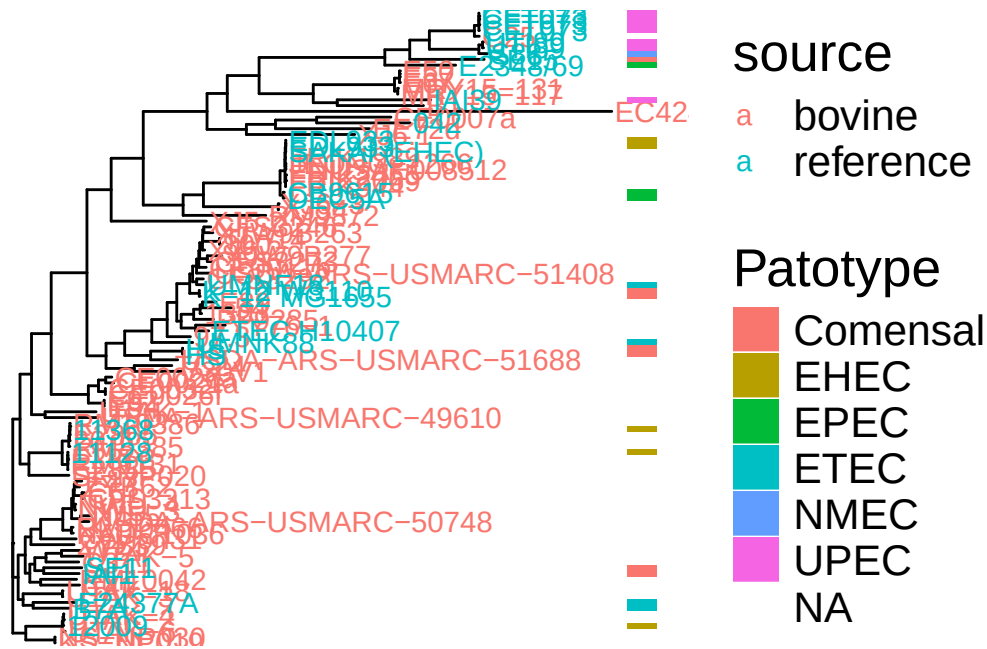
# The rows have to be in the same order as the tree
metadata_ordered <- metadata %>%
  filter(label %in% tree$tip.label) %>%
  arrange(match(label, tree$tip.label))

metadata_ordered$Phylogroup <- replace(metadata_ordered$Phylogroup, is.na(metadata_ordered$Phylogroup), "label")

metadata_ordered_2 <- metadata_ordered[,c(-1,-2)]
rownames(metadata_ordered_2) <- make.unique(metadata_ordered_2$Strain)
colnames(metadata_ordered_2)[1] <- "label"

# Change tip label
tree$tip.label <- metadata_ordered_2$label

# Plot tree with metadata
p <- ggtree(tree) %<+% metadata_ordered_2 +
  geom_tiplab(aes(color = source))
print(p)
```

Session Info

```
sessionInfo()
```

```
R version 4.4.2 (2024-10-31)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 22.04.5 LTS
```

Matrix products: default

```
BLAS:      /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
```

```
LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p-r0.3.20.so; LAPACK version 3.11.0
```

locale:

```
[1] LC_CTYPE=es_MX.UTF-8      LC_NUMERIC=C
[3] LC_TIME=es_MX.UTF-8      LC_COLLATE=es_MX.UTF-8
[5] LC_MONETARY=es_MX.UTF-8  LC_MESSAGES=es_MX.UTF-8
[7] LC_PAPER=es_MX.UTF-8     LC_NAME=C
[9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=es_MX.UTF-8 LC_IDENTIFICATION=C
```

```
time zone: America/Mexico_City
```

```
tzcode source: system (glibc)
```

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods   base
```

other attached packages:

```
[1] dplyr_1.1.4  ggplot2_3.5.1 treeio_1.28.0 ggtree_3.12.0
```

loaded via a namespace (and not attached):

```
[1] yulab.utils_0.1.8  utf8_1.2.4      generics_0.1.3   tidyr_1.3.1
[5] ggplotify_0.1.2    lattice_0.22-6  hms_1.1.3        digest_0.6.37
[9] magrittr_2.0.3     evaluate_1.0.1  grid_4.4.2       fastmap_1.2.0
[13] jsonlite_1.8.9     ape_5.8         tinytex_0.54     purrr_1.0.2
[17] fansi_1.0.6        aplot_0.2.4     scales_1.3.0     lazyeval_0.2.2
[21] cli_3.6.3          crayon_1.5.3    rlang_1.1.4      bit64_4.0.5
[25] munsell_0.5.1      tidytree_0.4.6  withr_3.0.2      yaml_2.3.10
[29] tools_4.4.2        parallel_4.4.2  tzdb_0.4.0       colorspace_2.1-1
[33] vctrs_0.6.5        R6_2.5.1        gridGraphics_0.5-1 lifecycle_1.0.4
[37] bit_4.0.5          fs_1.6.5        ggfun_0.1.8      vroom_1.6.5
[41] pkgconfig_2.0.3    pillar_1.9.0    gtable_0.3.6     glue_1.8.0
[45] Rcpp_1.0.13-1      xfun_0.49       tibble_3.2.1     tidyselect_1.2.1
[49] rstudioapi_0.17.1  knitr_1.49      farver_2.1.2     htmltools_0.5.8.1
[53] nlme_3.1-166       patchwork_1.3.0  rmarkdown_2.29   labeling_0.4.3
[57] readr_2.1.5        compiler_4.4.2
```